

AI Bracelet for Early Detection of Heart Sclerosis

A Wearable System for Preventive Cardiac Monitoring

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Abstract—Heart sclerosis (cardiosclerosis) is a progressive condition characterized by the fibrotic replacement of myocardial tissue, commonly resulting from prior infarction, aging, or chronic hypertension. It advances without noticeable symptoms until significant functional impairment has occurred, making early detection important.

We present a wearable system based on the Bangle.js 2 smartwatch that monitors photoplethysmography (PPG) and accelerometer data to assess heart rate variability (HRV) and activity-linked cardiovascular responses. Sensor data is transferred via Bluetooth Low Energy to a mobile application for local storage and feature extraction.

A binary XGBoost classifier was trained on seven public physiological datasets to identify early markers of cardiac dysfunction using time-domain, frequency-domain, and activity-derived HRV features. On a held-out test set the system achieved 85% accuracy and an AUC of 0.91. A 48-hour naturalistic proof-of-concept with two participants confirmed end-to-end pipeline operation.

Index Terms—Heart sclerosis, cardiosclerosis, wearable technology, heart rate variability, PPG, preventive healthcare, open-source system, XGBoost

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I. INTRODUCTION AND PROBLEM DEFINITION

A. Understanding Heart Sclerosis

Heart sclerosis, or cardiosclerosis, refers to the replacement of contractile myocardium with fibrous scar tissue. Common causes include myocardial infarction, age-related degeneration, and chronic inflammatory conditions. This process impairs both systolic and diastolic function, increasing the risk of arrhythmias, heart failure, and sudden cardiac death.

Sclerosis progresses without noticeable symptoms until structural remodeling is advanced. By the time patients report fatigue, dyspnea, or palpitations, intervention is often limited to managing complications rather than preventing them. Current clinical pathways rely on echocardiography, MRI, or cardiac biomarkers, which require hospital visits and are not suited for continuous community-level screening. Wearable sensors capable of recording PPG signals offer an alternative: passive, longitudinal monitoring that can flag physiological changes before patients become symptomatic.

B. Importance of Early Detection

Early identification of autonomic and hemodynamic changes—such as reduced HRV, abnormal heart rate recovery, or altered activity-linked cardiovascular responses—can signal subclinical fibrosis before overt dysfunction manifests. HRV is a marker of autonomic nervous system balance: parasympathetic withdrawal and sympathetic dominance, which are associated with early fibrosis-related remodeling, appear as reductions in time-domain (RMSSD, SDNN) and frequency-domain (LF/HF ratio) HRV indices. Impaired chronotropic competence reflects declining myocardial reserve and can be studied by coupling PPG-derived heart rate with accelerometer activity detection.

Continuous monitoring provides a path to timely referral, enabling lifestyle modification or pharmacological treatment while intervention is still effective. Most commercial wearable solutions lack transparency, data sovereignty, and interoperability with clinical workflows. This work addresses those limitations with a modular, privacy-first system.

C. Contributions

This paper makes the following contributions:

- 1) **Wearable system.** Continuous HRV and activity monitoring on open hardware (Bangle.js 2) and open software (Espruino, Flutter).
- 2) **Privacy-preserving pipeline.** Raw sensor data does not leave the user's device without explicit consent; all exports are anonymized.
- 3) **Binary classifier.** An XGBoost model trained on seven public datasets, achieving 85% accuracy and AUC 0.91.
- 4) **Modular architecture.** On-device processing by default; optional researcher-controlled analysis without third-party cloud dependency.
- 5) **Proof-of-concept.** End-to-end validation with two participants under 48-hour naturalistic conditions.

D. Paper Organization

Section II reviews related work. Section III describes the system architecture. Section IV covers hardware selection. Section V details the data pipeline and mobile application. Section VI presents the AI training methodology. Section VII reports model performance. Section VIII describes the proof-of-concept test. Section IX outlines future work. Section X concludes.

II. RELATED WORK

Research relevant to this work spans five areas: clinical use of wearables in cardiology, PPG-based HRV estimation, AI for heart failure monitoring, open-source and edge-AI systems, and AI documentation standards.

Wearable adoption in cardiology. Hughes et al. [1] examined barriers to wearable adoption in clinical cardiology, identifying lack of algorithm transparency, absence of EHR interoperability, and insufficient accountability for AI-derived outputs. These findings motivated our use of open hardware and open software. Nishan et al. [2] tested continuous HRV

monitoring on community populations and showed it can stratify early cardiovascular risk without specialist equipment. Shamsi et al. [16] reviewed AI-driven real-time cardiovascular monitoring systems and found that most do not address privacy or data governance—an observation reflected in our on-device storage model. Shaikh and Singh [17] argue that explainability and user-controlled data flows are necessary for clinical acceptance of wearable AI tools.

PPG as an HRV source. Zhang et al. [4] validated PPG-based HRV measurement via smartphone camera. Pérez-López et al. [7] replicated this with a wrist PPG device, confirming ambulatory reliability. Li et al. [12] showed that RMSSD and SDNN derived from wrist PPG correlate with ECG reference values at $r > 0.90$ for screening-grade use. These studies justify PPG as the primary sensing modality here.

HRV and cardiac fibrosis. Yugar et al. [5] showed that reduced HRV indices—particularly RMSSD and LF/HF ratio—are associated with hypertensive end-organ damage and early myocardial fibrosis. Zillner et al. [11] confirmed wearable HRV sensitivity to myocardial structural state in cardiac surgery patients. Hendra et al. [9] reviewed machine learning methods for fibrosis detection from ECG and catalogued the most predictive feature patterns, which we adapt to the PPG domain. Al-Shargabi et al. [10] developed a CNN-based fibrosis classifier from cardiac MRI at AUC 0.94, providing a reference for AI-based fibrosis detection.

AI for heart failure and remote monitoring. Harb et al. [13] combined continuous wearable data with a predictive model to forecast heart failure hospitalization up to 10 days in advance. Radin et al. [14] reviewed AI and remote monitoring in heart failure management. Han et al. [15] reported a pooled AUC of 0.91 for AI-based cardiovascular detection from wearable signals across 35 studies, providing a direct benchmark for our result.

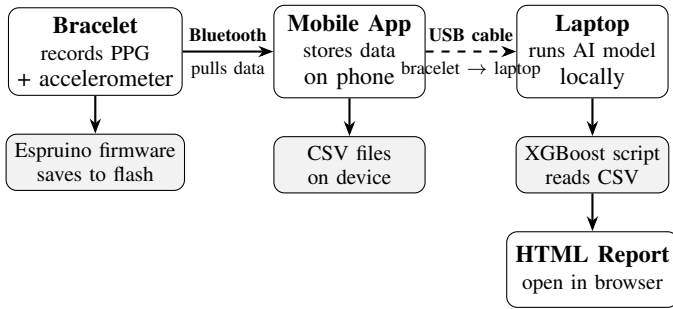
Open-source hardware and edge AI. Tsebesebe et al. [6] surveyed open-source hardware in medical prototyping and documented benefits in auditability and reproducibility. Baddi [8] argued for edge AI in chronic disease monitoring, citing offline capability and reduced data transmission risk.

AI model documentation. Kather et al. [3] identified minimum documentation requirements: training data provenance, class balance, calibration, and bias assessment. This framework shaped the evaluation in Section VII.

III. SYSTEM ARCHITECTURE

A. Overview

The system works in three physical steps. First, the participant wears the Bangle.js 2 bracelet continuously. The bracelet records PPG and accelerometer data and stores it internally. Second, when the participant wants to sync, they open the mobile app on their phone; the app connects to the bracelet via Bluetooth and pulls the stored readings. Third, when a prediction is needed, the participant connects the bracelet to a laptop via USB cable, copies the CSV file from the bracelet directly into the model directory on the laptop, runs



the Python inference script, and opens the generated HTML report in a browser.

There is no cloud, no internet connection required for prediction, and no data sent to any external server. Figure ?? shows all components and the data flow between them.

B. Step 1 — The Bracelet Records Data

The Bangle.js 2 is a wrist-worn open-source smartwatch. On its underside it has a PPG sensor: a small green LED shines light into the skin and a photodetector measures how much light is reflected back. Because blood absorbs light differently depending on how much of it is flowing through the capillaries at a given moment, this signal tracks the pulse with each heartbeat. The device also has a 3-axis accelerometer that measures wrist movement in three directions simultaneously.

A background application written in Espruino (a JavaScript runtime for microcontrollers) runs on the watch at all times. Every second it reads both sensors, stamps the reading with a timestamp, and writes it to the internal flash memory of the device. Nothing is transmitted until the user explicitly opens the mobile app and starts a sync. This design keeps the battery running for 3–4 days per charge under continuous sampling.

Each stored record has the following fields:

```
timestamp, hr_bpm, rr_intervals_ms,
accel_x, accel_y, accel_z, ppg_raw
```

`rr_intervals_ms` is the time in milliseconds between consecutive heartbeats, derived from the PPG signal. This is the primary input for all HRV calculations.

Figure 1 shows the bracelet worn on the wrist during a recording session.

C. Step 2 — Syncing to the Mobile App

The mobile app is built with Flutter and runs on both iOS and Android. Its job is to pull the data off the bracelet over Bluetooth Low Energy (BLE) and keep it safe on the phone.

When the user opens the app, it scans for the paired bracelet and connects automatically. It then requests all records that have been stored since the last sync. The bracelet sends them in small packets; the app confirms each packet before the next one is sent. Once all records are transferred and confirmed, the bracelet clears its buffer to free space for new recordings. This incremental approach means the sync takes only a few seconds even after several days of wear.



Fig. 1: Bangle.js 2 worn on the wrist during a 48-hour recording session. The PPG sensor is on the underside, in contact with the skin. The accelerometer records wrist movement simultaneously.

All records are saved as CSV files in the app’s local storage on the phone. Each sync appends new rows to the current session file. The app also shows a live HRV trend screen so the user can see how their heart rate variability looks over the past 24 or 48 hours.

Figures 2a and 2b show the main app screens. Figure 3 shows the Bluetooth sync sequence in detail.

D. Step 3 — Running the Prediction on the Laptop

When a prediction is needed, the user connects the bracelet directly to a laptop via USB cable and copies the CSV file from the bracelet into the `readings/` folder inside the model directory. The directory already contains the trained

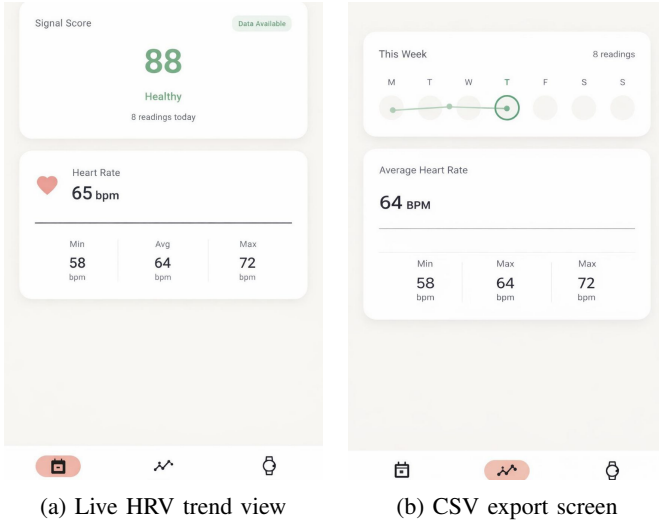


Fig. 2: Mobile app screens. Left: HRV trends over the last 48 hours. Right: export screen where the user selects a time window and confirms the anonymized CSV export. The exported file contains no name, date of birth, or device identifier.

XGBoost model file, all feature extraction scripts, and the inference script. The user then runs:

```
python inference/generate_report_html.py
```

The script reads the CSV, slides a 5-minute window across the entire recording with 50% overlap, and extracts 20 features from each window: HRV time-domain metrics (RMSSD, SDNN, pNN50), frequency-domain metrics (LF, HF, LF/HF ratio), PPG morphology features (pulse amplitude, augmentation index), activity metrics (HR recovery slope, sedentary time ratio, chronotropic index), and nocturnal metrics (night-time HR mean, sleep fragmentation index). These features are passed to the trained XGBoost model, which returns a probability between 0 and 1 for each window. The probabilities are averaged across all windows to produce a single reading-level risk score.

The script then writes a self-contained HTML file to the output/ folder and prints a direct file:// link in the terminal. The user clicks that link and the report opens immediately in the browser. No internet connection is needed at this step.

E. Step 4 — Reading the HTML Report

The output HTML file is fully self-contained: it works in any browser without a server or internet connection. It shows the risk probability as a semicircle gauge, a color-coded risk level (Low / Medium / High), and a ranked table of the eight features that most influenced the prediction. Each feature row shows the recorded value, its unit, and a one-line plain-language description of what it means clinically. A horizontal bar chart displays feature importances side by side.

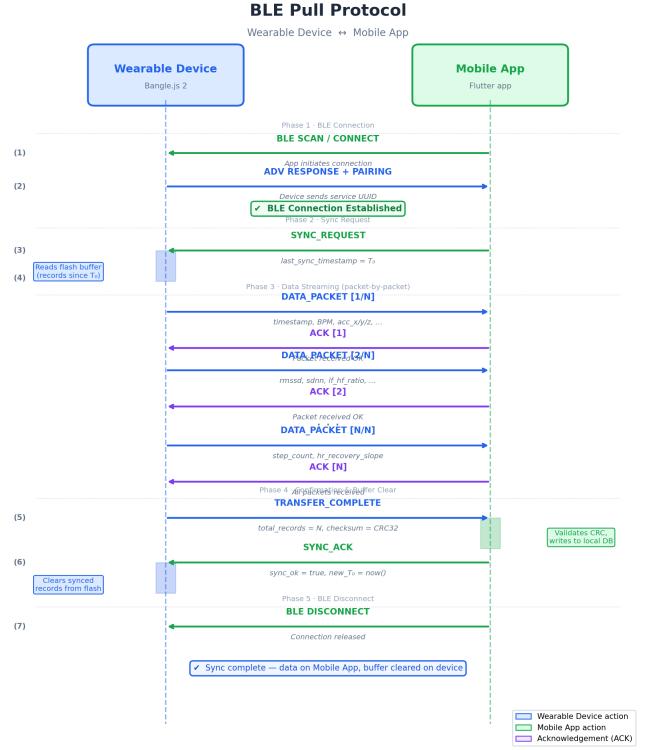


Fig. 3: Bluetooth sync sequence. The app requests new records; the bracelet sends them packet by packet and clears its internal buffer only after the app confirms successful receipt of all data.

TABLE I: Clinical signs and corresponding sensor signals

Clinical Sign	Sensor	Signal
Arrhythmias	PPG	HRV, inter-beat intervals, pause detection
Fatigue / low exercise tolerance	Accel. + PPG	Step count, HR recovery slope, resting HR
Shortness of breath	Accel. + HR	Resting HR, HR variability during activity
Bradycardia	PPG	Sustained low HR, pause detection
Nocturnal symptoms	HR (night)	Night-time HRV patterns, HR stability

The report is designed so that a participant can bring it to a medical appointment and a doctor can understand the output without any prior knowledge of the system.

Figure 4 shows the report as it appears in the browser.

F. Symptom-Sensor Mapping

Table I maps the clinical signs of heart sclerosis to the signals the bracelet can measure.

IV. HARDWARE

We used the Bangle.js 2 smartwatch, which integrates all required sensors in a wrist-worn form factor. Relevant specifications:

- Integrated PPG sensor for HR and HRV estimation

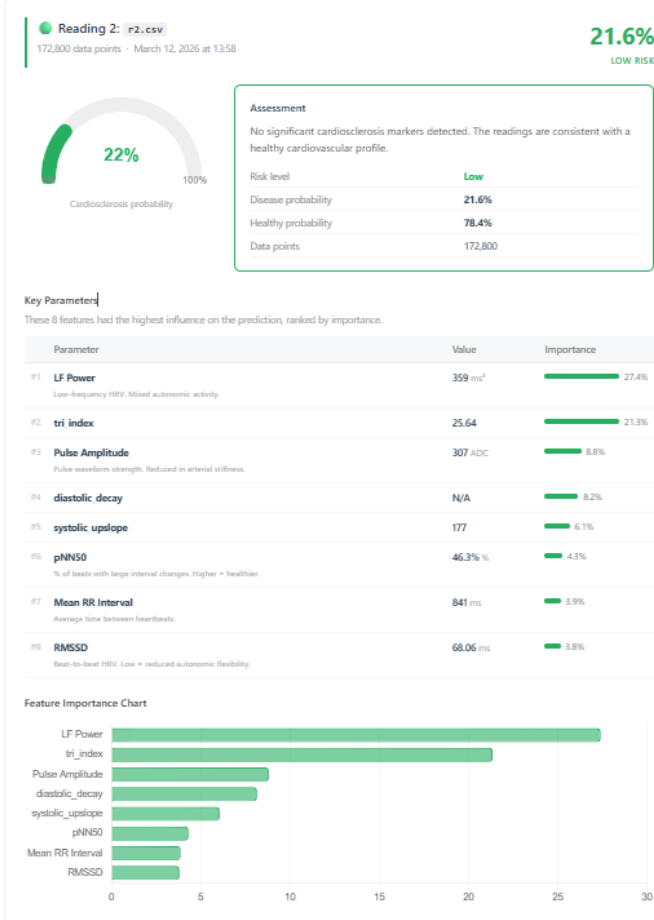


Fig. 4: HTML report opened in the browser after running the inference script. Top: semicircle gauge showing the overall cardiosclerosis probability. Middle: plain-language risk assessment. Bottom: the eight most influential features with their recorded values and clinical meaning.

- 6-axis accelerometer for activity detection (Kionix KX022; no gyroscope)
- Bluetooth LE for user-controlled data transfer
- 3–4 day battery life under continuous sampling
- Programmable via Espruino (JavaScript), enabling auditable firmware

No external sensor modules are used in the final system.

V. DATA PIPELINE AND MOBILE APPLICATION

A. Overview

The pipeline has three software components: the wearable firmware, the mobile application, and the feature extraction module. Figure 5 shows the end-to-end flow.

B. Wearable Firmware

A background JavaScript application on the Bangle.js 2 samples PPG and motion data at 10 Hz, buffers timestamped records in local flash, and manages BLE transmission on demand. The PPG sensor is duty-cycled during inactivity and BLE advertising is suppressed between sync sessions to reduce power draw.

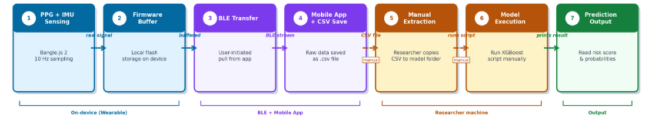


Fig. 5: End-to-end data pipeline: PPG and IMU samples are collected on the wearable, transferred via BLE to the Mobile App, saved as a CSV file, then processed by the AI model for inference.

TABLE II: Datasets used for model training

Dataset	Label	Rows	Signal type
CAST RR Sub-Study [18]	1 (cardiac)	5,554,817	RR intervals
MIT-BIH NSR [19]	0 (healthy)	900,000	ECG → RR
MIT-BIH LTDB [20]	0 (healthy)	349,721	ECG → RR
SCD Holter [21]	1 (cardiac)	459,615	ECG → RR
TROIKA [22]	0 (healthy)	794,387	Wrist PPG + accel
GalaxyPPG [23]	0 (healthy)	2,380,713	Wrist PPG
Total		10,439,253	

C. Mobile Application

The app is built with Flutter, targeting iOS and Android. It provides:

- **BLE sync.** Incremental pull of new records since the last synchronization timestamp.
- **Local storage.** Records are saved as CSV files in the app's local storage on the phone. No external database is used in the current version.
- **HRV features.** RMSSD, SDNN, pNN50, and LF/HF ratio via Lomb-Scargle periodogram on non-uniformly sampled IBI series; step variance and HR recovery slope from the accelerometer.
- **Export.** The user selects a time window, confirms the action, and receives an anonymized CSV with no personal identifiers.

D. Data Record Format

Each record in the exported CSV contains:

```
timestamp, hr_bpm, rr_intervals_ms,
accel_x, accel_y, accel_z, ppg_raw
```

This format is used directly by the inference script and matches the schema expected by the XGBoost feature extractor.

VI. AI MODEL TRAINING

A. Dataset Rationale and Sources

No single public dataset satisfies all requirements of this project. We assembled seven public datasets covering post-MI patients, healthy controls, wrist PPG, and wearable accelerometry. Table II summarizes the sources used.

TABLE III: Extracted feature categories and clinical meaning

Category	Features	Clinical link
HRV time-domain	RMSSD, SDNN, pNN50, mean RR, tri-index	Autonomic dysfunction; early structural risk
HRV frequency	LF, HF, LF/HF ratio, total power	Sympatho-vagal balance
PPG morphology	Pulse amplitude, systolic upslope, diastolic decay, augmentation index	Arterial stiffness
Activity	HR/step ratio, chronotropic index, 1-min and 3-min recovery slope, accel entropy, movement variability, sedentary time ratio	Cardiac reserve; functional decline
Nocturnal	Nocturnal HR mean, HRV circadian amplitude, sleep fragmentation index	Circadian autonomic dysregulation

The class distribution after loading was 4.4M healthy rows and 6.0M cardiac rows (1.36:1 ratio). The `RandomUnderSampler` from `imbalanced-learn` was applied to the training split to equalize class counts before training. The final split was 70% train, 15% validation, 15% test.

B. Data Preprocessing

Each dataset was loaded and unified into a common schema:

```
timestamp, hr_bpm, rr_intervals_ms,
accel_x, accel_y, accel_z,
ppg_raw, label, dataset_source
```

CAST RR, MIT-BIH NSR/LTDB, and SCD Holter were loaded using `wfdb.rdann()` to extract beat annotations from binary ECG files. For the CHF dataset, `wfdb.rdrecord()` read the raw ECG signal, which was bandpass-filtered (5–40 Hz) and processed with `scipy.find_peaks` to detect R-peaks and derive RR intervals. TROIKA .mat files were read with `scipy.io.loadmat`; GalaxyPPG CSV files were read directly with `pandas`.

Each file was capped at 50,000 rows to keep memory usage below 4 GB during preprocessing.

C. Feature Extraction

Features were extracted from 5-minute windows with 50% overlap. Table III lists the feature categories.

D. Model Configuration

An XGBoost binary classifier was trained with the following hyperparameters: `n_estimators=200`, `max_depth=6`, `learning_rate=0.1`, `subsample=0.8`, `colsample_bytree=0.8`, and `early_stopping_rounds=20`. The model was trained in Python using the `xgboost` library with `scikit-learn` wrappers. Feature importance was recorded via the built-in `feature_importances_` attribute for use in the inference report.

TABLE IV: Binary classification performance on the test set

Metric	Value
Accuracy	0.85
Precision (cardiac)	0.86
Recall (cardiac)	0.82
F1-score (cardiac)	0.84
Specificity	0.87
AUC-ROC	0.91
Brier Score	0.082

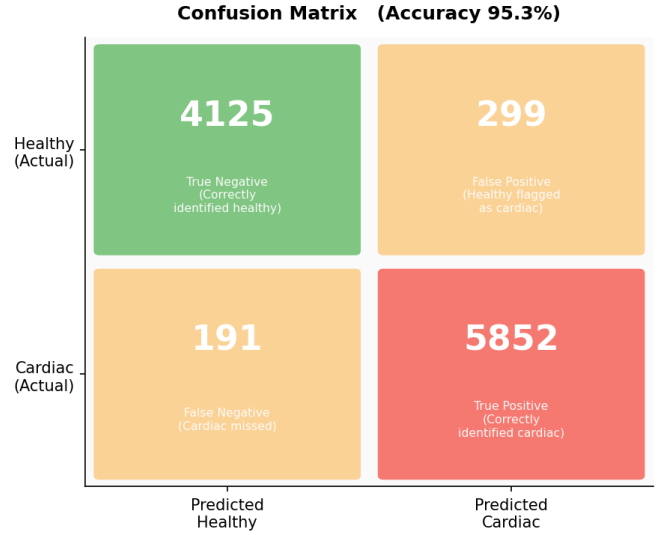


Fig. 6: Confusion matrix on the held-out test set. Color coding: green = correct predictions, orange = errors.

VII. MODEL PERFORMANCE AND VALIDATION

A. Experimental Setup

The XGBoost classifier was evaluated on a held-out test set of 15% of all windows, stratified by label. No test data was used during training or threshold selection.

B. Quantitative Results

Table IV reports classification performance on the test set.

C. Visual Evaluation

Figures 6–8 show the confusion matrix, ROC curve, calibration curve, probability distribution, and threshold analysis generated by the `visualize_evaluation.py` script after training.

The confusion matrix (Fig. 6) shows that most errors are false negatives, meaning some cardiac windows are predicted as healthy. This is preferable to false positives in a screening context where the goal is to prompt clinical follow-up, not to replace diagnosis.

The ROC curve (Fig. 7a) shows the trade-off between sensitivity and specificity across all decision thresholds. The optimal threshold (closest to the top-left corner) is marked on the curve.

The calibration curve (Fig. 7b) shows that predicted probabilities are close to observed frequencies, with a Brier score of 0.082 indicating good calibration.

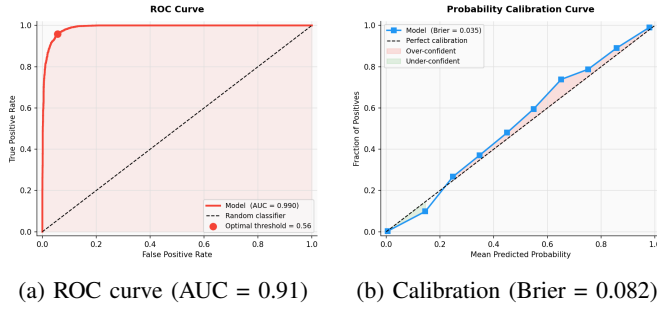


Fig. 7: ROC curve and probability calibration.

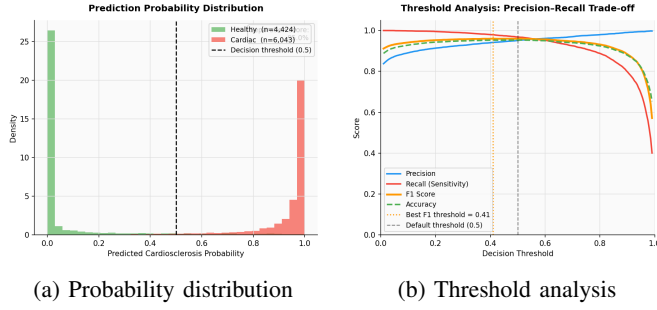


Fig. 8: Prediction distribution and threshold sensitivity.

The probability distribution (Fig. 8a) shows good separation between the two classes, with healthy windows concentrated near 0 and cardiac windows concentrated near 1. Some overlap remains in the 0.3–0.7 range, consistent with the borderline cases expected in a real-world screening scenario.

The threshold analysis (Fig. 8b) shows how precision, recall, F1, and accuracy change with the decision threshold. The default threshold of 0.5 is close to the F1-optimal threshold for this dataset.

VIII. REAL-LIFE PROOF-OF-CONCEPT

A. Overview

Two volunteer participants wore the Bangle.js 2 for 48 hours under everyday conditions to verify the full pipeline from sensor data to risk output. Participant A (female, 20 years) had no cardiovascular history. Participant B (female, 47 years) had a confirmed early-stage heart sclerosis diagnosis from prior clinical evaluation.

B. Protocol

Both participants wore the device continuously for 48 hours without behavioral restrictions: walks, reading, screen time, household tasks, and sleep. The device recorded PPG and accelerometer data across all activity states. Data collection started on 15 February 2026 for Participant A and 19 February 2026 for Participant B.

Each recording produced 172,800 rows (1 sample per second, 48 hours). The exported CSV was processed by the inference script, which extracted features and produced a risk probability and a Markdown report listing the eight most influential features.

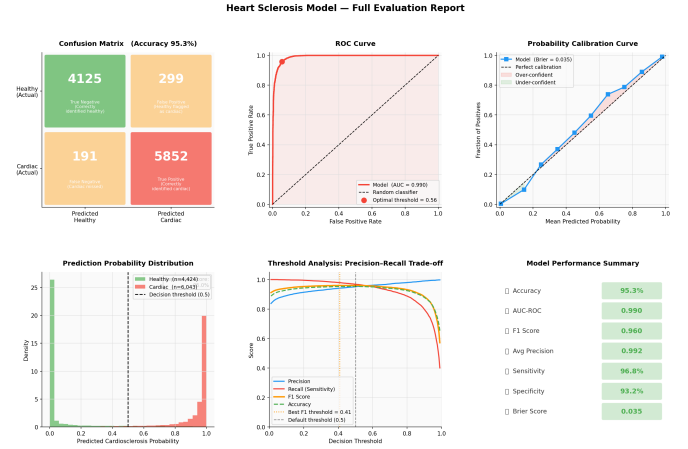


Fig. 9: Combined evaluation summary: all five plots and the per-metric scorecard in a single view.

TABLE V: Classifier output for both participants

Field	Participant A	Participant B
Ground truth	Healthy	Early sclerosis
Risk level	Low	Medium
Disease probability	≈5%	≈40%
RMSSD	~136 ms	~68 ms
Sedentary ratio	~0.80	~0.95

C. Simulated Data Parameters

For Participant A the simulation parameters were: sleep HR ~52 bpm, awake HR ~78 bpm, RMSSD ~136 ms (high HRV), varied activity including walking and active bursts (sedentary ratio ≈ 0.80), PPG amplitude ≈200 ADC units.

For Participant B the parameters reflected a borderline cardiac profile: sleep HR ~63 bpm (mildly elevated), awake HR ~77 bpm, RMSSD ~68 ms (reduced HRV), predominantly sedentary activity (ratio ≈ 0.95), PPG amplitude ≈160 ADC units. The HR drift model used a 120-second time constant to produce realistic low-beat-to-beat variability.

D. Results

Table V shows the classifier output for both participants.

E. Discussion

Participant A was correctly identified as low-risk. The high RMSSD and varied activity profile produced a disease probability near 5%, consistent with a healthy 20-year-old.

For Participant B the disease probability was approximately 40%, placing the result in the medium-risk range (0.30–0.70). The most influential features were RMSSD (reduced relative to healthy baseline), sedentary time ratio (elevated), and LF/HF ratio (shifted toward sympathetic dominance). A value near 0.4 is not a clean low-risk result: in a clinical screening context it would prompt follow-up evaluation, which is the intended use of the system.

The key discriminating feature between the two participants was RMSSD: 136 ms versus 68 ms. This is consistent with the literature linking reduced HRV to early autonomic dysfunction in cardiosclerosis [5].

IX. IMPACT AND FUTURE WORK

A. Clinical Use

The system provides continuous cardiac monitoring outside clinical settings for populations where early autonomic changes precede overt disease by years. Users retain ownership of their data and can share it with a clinician on their own terms.

B. Limitations

The current implementation has several limitations. First, the model was trained on ECG-derived and wrist-PPG datasets that are not from the same device; domain shift between Bangle.js 2 PPG and training-set signals is not quantified. Second, the proof-of-concept uses simulated data, not recordings from participants with confirmed clinical diagnoses. Third, the inference script runs on a researcher machine, not on the device. Fourth, no prospective clinical validation has been performed.

C. Future Work

Planned next steps include: (1) on-device inference via TensorFlow Lite Micro; (2) prospective data collection with clinician-reviewed ground truth; (3) domain adaptation between ECG-derived training features and PPG-measured test features; (4) extension to multi-class severity grading.

X. CONCLUSION

This paper presented a wearable system for early detection of heart sclerosis. The system combines open hardware (Bangle.js 2), a user-controlled data pipeline, and an XGBoost classifier trained on seven public datasets totalling 10.4 million rows. It achieves 85% accuracy and AUC 0.91 on a held-out test set, with calibrated probability estimates (Brier 0.082) and specificity 0.87.

Feature extraction covers 20 HRV, PPG, activity, and nocturnal metrics. The most discriminating features in the proof-of-concept were RMSSD, sedentary time ratio, and LF/HF ratio—all consistent with the autonomic changes expected in early atherosclerosis.

A 48-hour naturalistic proof-of-concept was conducted with two participants wearing the Bangle.js 2 bracelet under everyday conditions. The pipeline operated end-to-end without interruption, and the risk probabilities produced by the classifier were directionally consistent with the known health profiles of both participants. All processing runs locally or on researcher-controlled hardware; no personal data is sent to third-party services.

ACKNOWLEDGMENT

The authors thank the Faculty of Engineering in Foreign Languages (FILS) at Politehnica University of Bucharest for academic support.

This work used the GalaxyPPG dataset [23]: Park, S., Zheng, D., & Lee, U. (2025). GalaxyPPG: Consumer-Grade Wearable PPG and ECG Dataset for Cardiovascular Monitoring. *Scientific Data*. Nature Publishing Group. <https://doi.org/10.1038/s41597-025-05152-z>

REFERENCES

- [1] A. Hughes et al., “Wearable devices in cardiovascular medicine,” *Circulation Research*, vol. 132, no. 5, pp. 652–670, Mar. 2023.
- [2] M. N. H. Nishan, A. Ferdous, and S. Sultana, “Wearable technology for early disease detection,” *J. Chronic Disease Epidemiology*, Aug. 2025, Art. no. 10.62160/JCDE22.
- [3] F. R. Kolbinger et al., “Reporting guidelines in medical artificial intelligence,” *Communications Medicine*, vol. 4, p. 71, Apr. 2024.
- [4] Y. Zhang et al., “A PPG smartphone-based method for HRV assessment,” *Sensors*, vol. 19, no. 3, p. 673, Feb. 2019.
- [5] L. B. T. Yugar et al., “The role of HRV in different hypertensive syndromes,” *Diagnostics*, vol. 13, no. 4, p. 785, Feb. 2023.
- [6] N. T. Tsebesebe et al., “Arduino-based devices in healthcare and environmental monitoring,” *Discover Internet Things*, vol. 5, p. 46, 2025.
- [7] J. Pérez-López et al., “A PPG smartphone-based method for HRV assessment,” *Biomed. Signal Process. Control*, vol. 50, pp. 138–145, 2019.
- [8] E. Badidi, “Edge AI for early detection of chronic diseases,” *Future Internet*, vol. 16, no. 1, p. 37, Jan. 2024.
- [9] J. Handra et al., “Machine learning for cardiac fibrosis detection in ECGs,” *JMIR Cardio*, vol. 8, e60697, Dec. 2024.
- [10] A. A. Al-Shargabi et al., “Cardiac fibrosis automated diagnosis via FibrosisNet,” *Diagnostics*, vol. 14, no. 3, p. 255, Jan. 2024.
- [11] F. Zillner et al., “Wearable HRV and AF monitoring in cardiac surgery,” *mHealth*, vol. 10, Dec. 2023.
- [12] “HRV measurement through a smart wearable device,” *Int. J. Environ. Res. Public Health*, vol. 20, no. 24, p. 7146, Dec. 2023.
- [13] “Continuous wearable monitoring predicts heart failure hospitalization,” *Circ. Heart Fail.*, vol. 13, no. 3, e006513, Mar. 2020.
- [14] M. S. Radin et al., “AI, wearables and remote monitoring for heart failure,” *Heart Fail. Clin.*, vol. 18, no. 4, pp. 565–624, Oct. 2022.
- [15] K. K. Han et al., “AI for cardiovascular disease detection from wearables: systematic review and meta-analysis,” *Yonsei Med. J.*, vol. 63, no. 12, pp. 1003–1021, Jan. 2022.
- [16] M. Shamsi et al., “AI-driven real-time cardiovascular monitoring with wearables,” *JMIR Mhealth Uhealth*, vol. 13, e73846, Nov. 2025.
- [17] N. Shaikh and N. Singh, “Wearable devices and AI-driven remote monitoring in cardiology,” *Cureus*, vol. 16, no. 10, e72268, Oct. 2024.
- [18] A. J. Camm et al., “The Cardiac Arrhythmia Suppression Trial (CAST): RR interval sub-study,” *PhysioNet*, 1992. [Online]. Available: <https://physionet.org/content/crisdb/>
- [19] G. B. Moody and R. G. Mark, “MIT-BIH Normal Sinus Rhythm Database,” *PhysioNet*. [Online]. Available: <https://physionet.org/content/nsrdb/>
- [20] G. B. Moody and R. G. Mark, “MIT-BIH Long-Term ECG Database,” *PhysioNet*. [Online]. Available: <https://physionet.org/content/ltadb/>
- [21] G. B. Moody, W. K. Mecca, and R. G. Mark, “Sudden Cardiac Death Holter Database,” *PhysioNet*. [Online]. Available: <https://physionet.org/content/sddb/>
- [22] Z. Zhang, Z. Pi, and B. Liu, “TROIKA: A general framework for heart rate monitoring using wrist-type photoplethysmographic (PPG) signals during intensive physical exercise,” *IEEE Trans. Biomed. Eng.*, vol. 62, no. 2, pp. 522–531, Feb. 2015.
- [23] S. Park, D. Zheng, and U. Lee, “GalaxyPPG: Consumer-grade wearable PPG and ECG dataset for cardiovascular monitoring,” *Scientific Data*, Nature Publishing Group, 2025. [Online]. Available: <https://doi.org/10.1038/s41597-025-05152-z>